BMCS 4480
Fall 2026

Assignment 2

Submission Due: **October 11th, 11:59 pm**

Submit your work in the form of an executed Python Jupyter notebook (preferred), or Python/MATLAB/R code and writeup containing figures (pdf) to Courseworks. Existing libraries such as Scanpy, Seurat, and Scikit-Learn can be used, but we encourage implementing your own functionalities where possible. Please mention the names of all collaborators you may have discussed these problems with. Usage of AI is only permitted for debugging and searching references for library documentation.

Download single-cell gene expression data for ~6K PBMCs (peripheral blood cells) from a healthy donor from here. Note the data is already filtered and contains only the 500 most variable genes. Rows and columns represent genes and cell IDs respectively.

Q1a. Normalize the data by scaling to median library size and log transform the normalized data and perform PCA, followed by t-SNE or UMAP on the top 20 PCs. Plot the 2D or 3D embedding [10 pts].
Q1b. Is 20 a good choice for the number of PCs? [bonus 5 pts]

Q2. Cluster cells using the K-means method and color the embedding from Q1 with cluster IDs. Justify your choice of K [15 pts].

Q3. Compute a 30-NN (nearest neighbor) graph between cells. Plot a heatmap of the adjacency matrix for the graph [10 pts]. Justify your distance metric [5 pts].

Q4. Cluster cells using a graph-based algorithm such as Louvain with the kNN graph from Q3. Color the embedding with cluster IDs. How does it compare to K-means? [15 pts]

Q5a. Perform a t-test to find differentially expressed genes (DEGs) in a cluster of your choice. Color the embedding with the expression of the top 5 DEGs [15 pts].
Q5b. Based on the identified DEGs how would you characterize the cluster [5 pts]?

Q5c. Which other differential expression method might be appropriate for the expression distribution in this data? [bonus 5pt]

Q6. In a simplified model, four adjacent bins on one chromosome have hidden states $Z_t \in \{N, A\}$ (neutral or amplified) and observed expression categories $X_t \in \{L, H\}$ (low or high). Note this is a simplified model, not a complete copy number variation (CNV)-inference method. Initial probabilities are (0.5,0.5). Rows denote the previous state and columns the next state, in the order (N,A):

$$T = \begin{pmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{pmatrix}.$$

Emission probabilities are $P(H|N) = 0.2$ and $P(H|A) = 0.8$ with the complementary probabilities for L.

Q6a. Write the joint distribution of the hidden states and observations. State the transition and emission conditional-independence assumptions. [5 pt]

Q6b. Calculate the joint probability of the observations (H,H,L,H) with each of the hidden-state sequences (A,A,A,A) and (A,A,N,A). Show every transition and emission factor. Which sequence has the larger probability, and discuss if/why can a low-expression observation be assigned to an amplified state? Explain whether your calculated probabilities are joint or posterior probabilities. [10 pt]

Q6c. A colleague proposes sorting the genomic bins by observed expression before applying the HMM. Discuss how this would change the results [5 pt]

Q6d. Replace both diagonal entries of the transition matrix by (q) and both off-diagonal entries by (1-q), where (0<q<1). Derive the range of (q) for which (A,A,A,A) has a larger joint probability than (A,A,N,A). Compare only these two candidate sequences. Interpret the threshold biologically. [bonus 5 pt]

Q7. Solve problem 9.4 of the PRML textbook (pg 456) [5 pts]. Hint: You do not need to analytically prove the E step. Use 9.29 and 9.30 to create your own Q' with Bayes' rule for the M step, don't start with 9.30.